



Immigration Policy Agreement Moderated through Social Media

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Research Question & Hypothesis

Question: Does exposure to public approval of immigration policies on social media cause individuals to express greater approval of these policies?

Hypothesis: Exposure to high levels of support for immigration policies on social media leads individuals to conform by self-reporting higher levels of agreement with these policies.

Experimental Design



Randomization Scheme

- Partnered with PureSpectrum for Stratified Random Sample (US Census):
 - Political Party
 - Race
 - Hispanic / Latino Origin
 - Gender + Age Joint Distribution
- Used Qualtrics Quotas: random assignment & ensure blocking on party + ethnicity
- Sample Size = 200, 50-50 Split

▼ Current respondent breakdown

☐ Show quota labels ⓘ

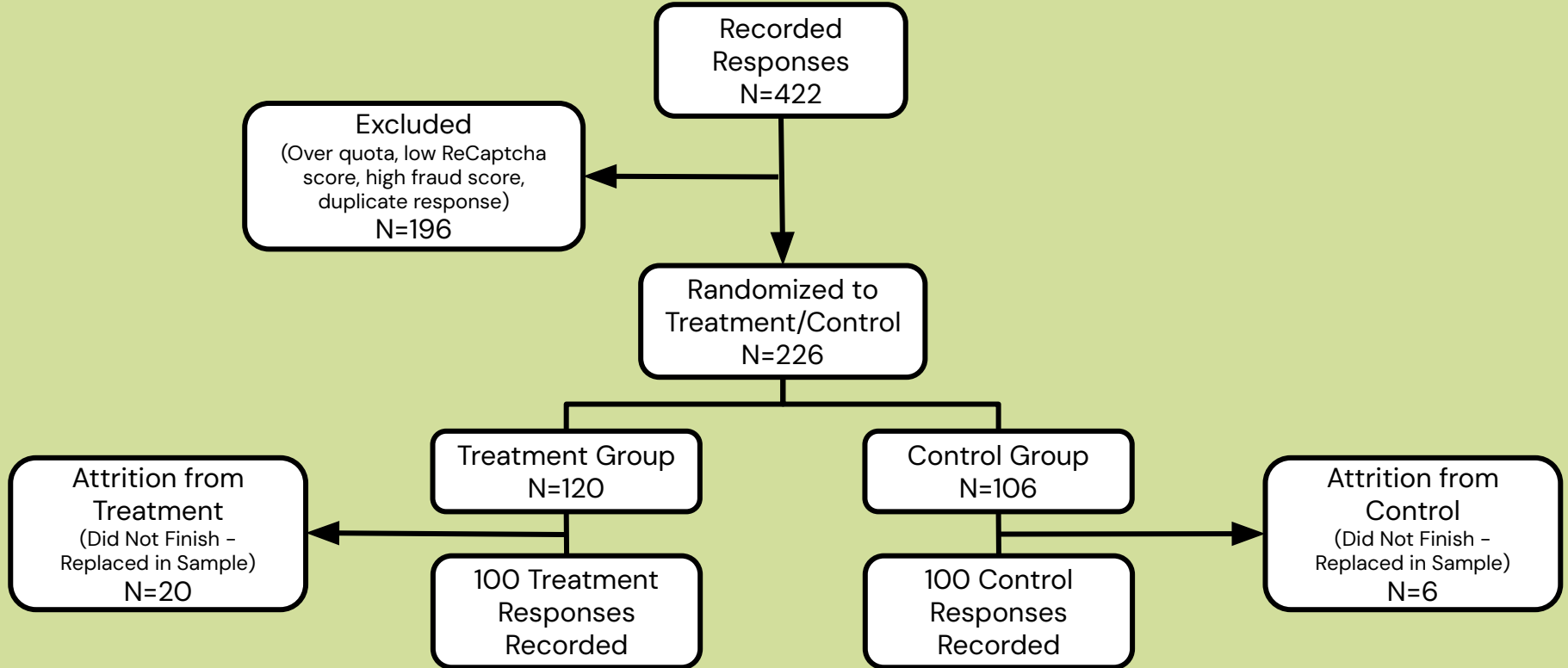
Quota description	Allocation %	Quota counts
Control is equal to True AND NQ2.9 Are you Hispanic or Latino? Yes is Selected	9.35%	15 / 19 ***
Control is equal to True AND NQ2.9 Are you Hispanic or Latino? No is Selected	40.65%	84 / 83 ***
Treatment is equal to True AND NQ2.9 Are you Hispanic or Latino? Yes is Selected	9.35%	15 / 19 ***
Treatment is equal to True AND NQ2.9 Are you Hispanic or Latino? No is Selected	40.65%	80 / 83 ***

Attrition & Balance Tests

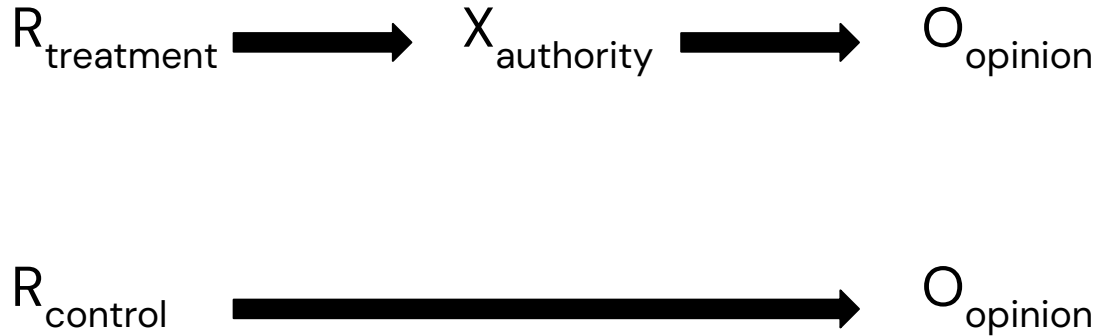
- Covariate balance is enforced by blocking approach
- Evidence of higher attrition in the Treatment group (t-test p-value: 0.009)
- No other covariates predict attrition
- Attrition not a statistically significant predictor for first treatment response

Randomized Group	Incomplete Surveys	Complete Surveys	Total Surveys
Control	6	100	106
Treatment	20	100	120

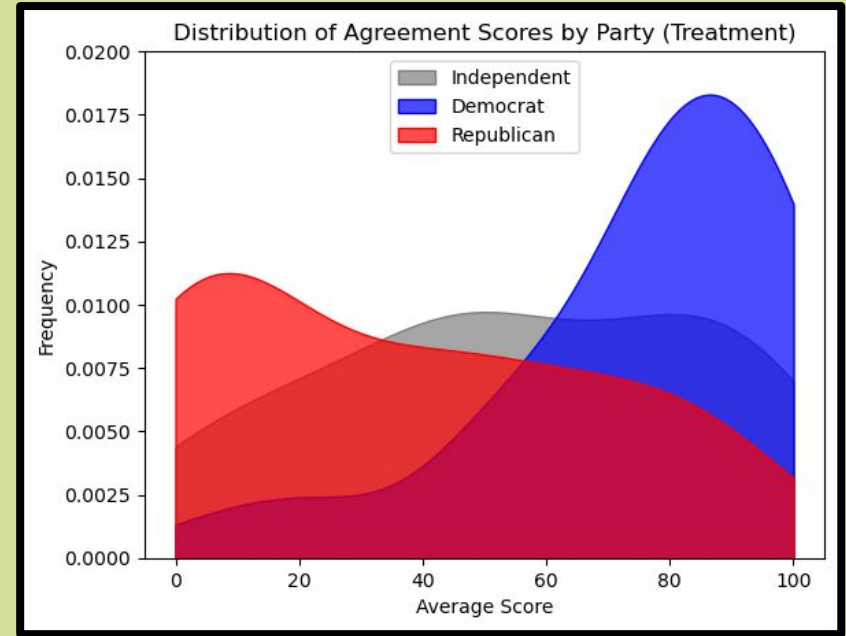
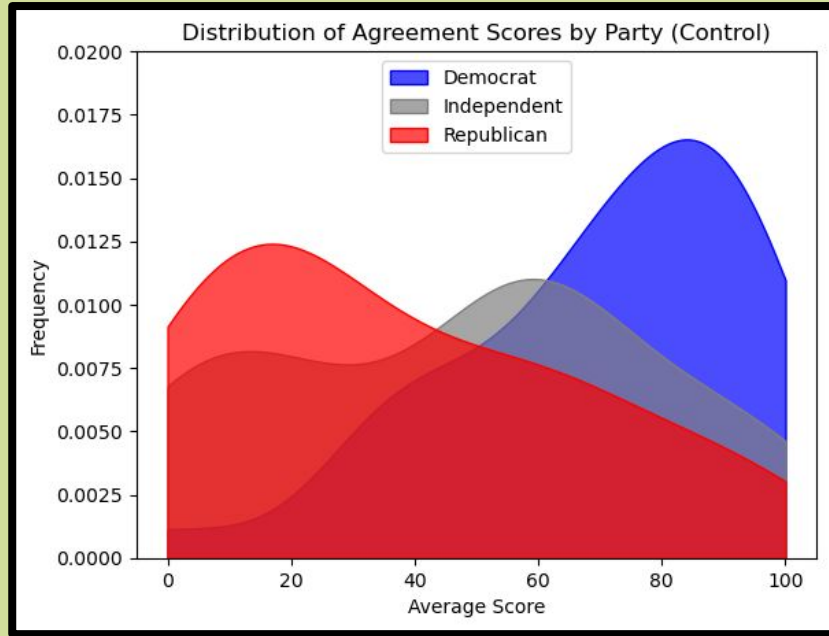
Flowchart



Posttest Only Randomized Experiment



Target Variable Breakdown



Results

Basic Model: R-squared 0.411, Adj. R-squared 0.323

	coef	std err	z	P> z	[0.025	0.975]
const	84.3415	12.487	6.754	0.000	59.868	108.815
is_treatment	5.5953	4.331	1.292	0.196	-2.894	14.084
party_Independent	-16.0286	6.534	-2.453	0.014	-28.834	-3.223
party_Other	-19.2193	416.143	-0.046	0.963	-834.844	796.406
party_Republican	-19.6130	7.145	-2.745	0.006	-33.617	-5.609
leaning_Conservative	-46.8587	8.638	-5.425	0.000	-63.789	-29.928
leaning_Lean Conservative	-35.2067	11.312	-3.112	0.002	-57.377	-13.036
leaning_Lean Liberal	-17.3658	8.861	-1.960	0.050	-34.734	0.002
leaning_Liberal	-13.3913	7.745	-1.729	0.084	-28.572	1.789
leaning_Moderate	-17.8222	6.654	-2.679	0.007	-30.863	-4.781
leaning_Strongly Conservative	-34.7083	10.713	-3.240	0.001	-55.705	-13.712
gender_woman	-3.4672	4.584	-0.756	0.449	-12.452	5.518
age_25-34	7.2072	8.758	0.823	0.411	-9.959	24.373
age_35-44	2.0478	9.013	0.227	0.820	-15.617	19.712
age_45-54	2.6571	9.503	0.280	0.780	-15.969	21.284
age_55-64	0.5171	9.592	0.054	0.957	-18.283	19.317
age_65 or older	-6.6510	9.006	-0.738	0.460	-24.303	11.001
race_American Indian or Alaska Native	4.8878	14.937	0.327	0.744	-24.389	34.164
race_Asian	4.0313	9.960	0.405	0.686	-15.489	23.552
race_Black or African American	0.0547	7.661	0.007	0.994	-14.961	15.071
race_Native Hawaiian or Other Pacific Islander	11.0322	176.154	0.063	0.950	-334.224	356.289
race_Other race	5.4656	9.934	0.550	0.582	-14.004	24.935
race_Two or more	4.7455	10.526	0.451	0.652	-15.886	25.377
hispanic_Yes	-1.7514	8.275	-0.212	0.832	-17.971	14.468
citizenship_Lawful Permanent Resident	27.6849	18.802	1.472	0.141	-9.167	64.536
citizenship_Naturalized U.S. citizen	-12.2263	11.244	-1.087	0.277	-34.263	9.811
citizenship_Not a U.S. citizen	18.4289	76.200	0.242	0.809	-130.921	167.779

Baseline Variables:

Party: Democrat
 Leaning: Strongly liberal
 Gender: Man
 Age: 18-24
 Race: White
 Hispanic: No
 Citizenship: U.S. Citizen

Results Contd.

Heterogeneous treatment effects are **not** significant and do **not** pass the likelihood ratio test

	coef	std err	z	P> z	[0.025	0.975]
Democrat_x_treatment	6.1832	5.588	1.106	0.269	-4.770	17.136
Republican_x_treatment	-1.8775	6.421	-0.292	0.770	-14.463	10.708
Independent_x_treatment	5.0448	7.891	0.639	0.523	-10.422	20.512
Other_x_treatment	-6.7124	5.663	-1.185	0.236	-17.811	4.387

Survey Question	ATE	P value
1	4.38	0.359
2	6.32	0.196
3	7.58	0.123
4	6.05	0.228
5	3.64	0.475

While a positive ATE is recorded for each question individually no effect is statistically significant from 0

Analysis & Learnings

- Even in the best fitting model the R-squared of 0.411 underscores the amount of noise present in the data. Of the covariates only “party” and “leaning” held significant predictive power meaning other covariates were ineffective at improving efficiency.
- Other covariates that should be considered in a future larger study are income, education level, political engagement and population density of residence.
- In the initial power analysis finding significant results was highly dependent on homogeneous treatment effects contrary to the results recorded.
- In pilot study it was pointed out that the fake posts could at times be recognized as fake, despite revisions this still may have played a factor.
- In research review most other studies moderated conformity through messages from specific political persons and parties which may play a larger role than peers.
- In research ATE's were typically lower in digital settings as well.

Open Questions

- Is a homogeneous treatment effect expected in a social media context? Is enabling political polarization inherent to the medium?
- What components of social media leads to an effect if any?
- What responsibility do social media companies have if people are affected by their content?